AIP Conference Proceedings

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Cite as: AIP Conference Proceedings **2670**, 030003 (2022); https://doi.org/10.1063/5.0115314
Published Online: 05 December 2022

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Methods to Solve Forward Kinematics of Parallel and Serial Manipulators

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Abstract. In this paper, approaches for the solution of parallel manipulators forward kinematics have been addressed. Current research on this topic has been analyzed to determine the status and unresolved issues of the research field. Articles on parallel manipulators forward kinematics were reviewed, and numerous approaches for resolving forward kinematics in particular have been identified and published, which may assist researchers in selecting the most appropriate technique for solving the problem in this domain. Researchers have also attempted to apply these methods to a variety of other parallel manipulator-related problems, such as inverse kinematics and dynamics. In robotics, serial manipulators were invented before parallel manipulators. Hence, methodologies for solving forward kinematics were first developed for serial manipulators and many methods are ineffective for parallel manipulators. In this paper, all identified methods can be used for solving forward kinematics of parallel and serial manipulators.

Keywords: *Parallel Manipulator, Forward Kinematics, Inverse Kinematics, Degrees-of-freedom, Screw Theory*

INTRODUCTION

Parallel robots have a long history, dating back to 1928, when James E. Gwinnet created a spherical parallel robot for a movie theatre. This invention was not well-known at the time. Willard L.V Pollard built the first industrial parallel robot for autonomous spray painting after ten years. Dr. Eric Gough developed a six d-o-f parallel robot (Octahedral Hexapod) for tyre testing in 1947, which brought a new era of robots. A universal rig was the name given to it. The Stewart platform, which was built in 1965, transformed the octahedral hexapod. This prompted plenty of ideas from researchers, which are eventually adopted by academics. Parallel robots are used, in a variety of applications including medical, space, entertainment and other domestic and military applications [1].

Parallel robot applications include simulators, micro-motion manipulators, machine tools, finishing operations, grasping, assembly and industrial applications. These robots can be utilized as pointing devices, walking machines, mobile robots, underwater robots, biologically inspired robots, and humanoid robots, among other things [2].

Closed-loop manipulator with a closed-loop kinematic chain is known as parallel manipulator (PM). It has a fixed base platform and a moving platform. The end-effector would be the moving platform, which would be connected to the fixed platform by multiple kinematic chains. Due to their superior stiffness, rigidity, mobility, precision, load carrying capability, and architecture, parallel robots are becoming increasingly popular in all disciplines of technology. These characteristics set them apart from ordinary serial robots.[3].

Kinematics is the study of relative motion between the manipulator's various linkages. During the study of kinematics, two types of issues arise. Firstly, the position analysis problem, and two the velocity analysis problem. Direct position and inverse position problems are the two types of position analysis problems. Similarly, there is a direct velocity problem and an inverse velocity problem in the velocity analysis problem. Forward Kinematics (FK)

International Conference on Robotics, Automation and Intelligent Systems (ICRAINS 21)
AIP Conf. Proc. 2670, 030003-1–030003-7; https://doi.org/10.1063/5.0115314
Published by AIP Publishing. 978-0-7354-4287-0/$30.00

is another name for direct kinematics. FK calculates the end-effector position from the given joint positions using the robot's kinematic equation. Inverse Kinematics (IK) is the opposite of FK [3].
In this paper, section 2 focuses on various methods used to solve FK problem. Table 1 shows the advantage and disadvantage of each method.

METHODS

Denavit-Hartenberg (D-H) Method

If a robot's kinematic linkages have either rotational joints or translational joints between them, a combined mathematical representation of rotational and translational displacements is required. A transformation matrix can be used to represent these motions. Only when the manipulator links are joined by rotary (R) or prismatic (P) joints can Denavit-Hartenberg (D-H) Method be used. The four D-H parameters are twist angle (β), link length (l), joint angle (ϕ), and joint offset (j). The motion between any two links considered at the same time will be defined by these characteristics. The D-H transformation matrix between two links is given in Eq. (1)[2]

$$T=\begin{bmatrix} \cos\phi & -\cos\beta\sin\phi & \sin\beta\sin\phi & l\cos\phi \\ \sin\phi & \cos\beta\cos\phi & -\sin\beta\cos\phi & l\sin\phi \\ 0 & \sin\beta & \cos\beta & j \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1}$$

The D-H transformation matrix is a 4×4 matrix made up of four sub-matrices. The orientation between moving frame and fixed frame is represented by the 3×3 sub-matrix (upper left). The 3×1 sub-matrix (top right) represents the moving frame's position with relation to the stationary frame. The perspective transformation is represented by the 1×3 sub-matrix (lower left), while the scaling factor is represented by the 1×1 sub-matrix (lower right). The overall matrix shows the end-effector's position in relation to the fixed base frame attached to the fixed link [2]. Other than the D-H transformation matrix, there are several forms of transformation matrices. A transformation matrix relates a moving coordinate frame to a fixed coordinate frame. It is possible to calculate the position and orientation of the moving frame in this manner.

T. K. Tanev [4] used homogeneous transformation matrix to solve FK.. Transformation matrix was developed to solve FK for a DELTA-Redundant robot having 3 d-o-f [5] and for a novel 3 d-o-f uncoupled spatial PM [6].

Screw Theory (ST)

Screw theory is a tool for mathematicians. A six-dimensional vector consisting of two 3D vectors is called as screw. This method was created by Sir Robert Stawell Ball to solve the kinematics and statics of rigid bodies. It is given as; ;$\$ = \begin{bmatrix} \hat{s} \\ \vec{s}_o \end{bmatrix}$. It is divided into two parts: primal and dual. The initial part ($\hat{s}$) is a unit vector, while the duplex part (s_O) is the moment of that unit vector around a fixed point as shown in Fig. 1. This fixed point is on the reference frame.

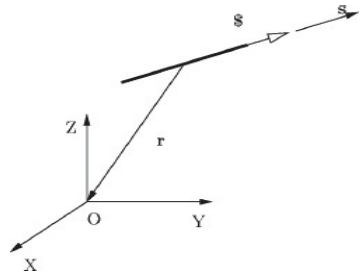

FIGURE 1. Screw

Moment is calculated using the formula $\overrightarrow{\mathbf{s_o}} = \hat{\mathbf{s}} \times \overrightarrow{\mathbf{r_o}} + \mathbf{h}\hat{\mathbf{s}}$, , where 'h' represents pitch of the screw and $\overrightarrow{r_o}$ represents a vector. As a result, the moment is calculated using the screw pitch and the vector. As shown in Fig. 2, screw's pitch is defined by its rotation and translation in same axis.

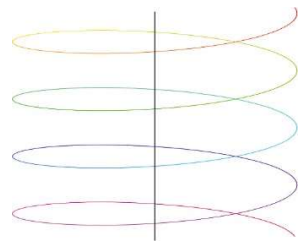

FIGURE 2. Pitch of a screw

Twist or velocity state can be found out by;$V = \begin{bmatrix} \overrightarrow{\omega} \\ \overrightarrow{v_o} \end{bmatrix}$, where, '$\overrightarrow{\omega}$' represents angular velocity and '$\overrightarrow{v_o}$' represents velocity of point 'o' on the rigid body. This point coincides instantly with the fixed point of the reference frame. Twist is 6D vector.

Similarly, acceleration state is given by$\overrightarrow{A_R}$= $\begin{bmatrix} \dot{\overrightarrow{\omega}} \\ \overrightarrow{a_o} \end{bmatrix}$, where, '$\dot{\overrightarrow{\omega}}$' is angular acceleration and '$\overrightarrow{a_o}$' represents point 'o' acceleration.

Different researchers tried to adopt screw theory. Duffy [7] analyzed the instantaneous kinematics of the planar serial and parallel mechanism using screw theory. The method of synthesizing Stewart platform was explored by Wolf & Soham[8] with Screw theory. Kim et al. [9] concentrated on a parallel planar robot's graphical instantaneous center using screws theory.

When describing the motion, screw is referred to as a twist, and when describing the motion and force of a rigid body, it is referred to as a wrench. The rigid body does no work when a wrench acts on it during an infinitesimal twist, hence these two screws are said to be reciprocal screws. The intuitive geometric approach was explained by Zhao et al. [10] using screw theory. The reciprocal screw theory was used by Mustafa & Agrawal [11] to study the force closure of cable-driven open system.

Euler Angle Representation (EAR)

A rotation matrix in terms of Euler angles about an axis can be constructed when a rigid body rotates around any of the three axes (x, y, or z-axis) [2]. Because rotation also indicates motion, a rotation matrix with three independent parameters represent motion. The orientation can be represented using these three Euler angle parameters. A moving coordinate frame and fixed coordinate frame connected to the ends of a rigid link can be used to create this representation. When a rigid body performs a rotation about z-axis, the rotation matrix ${}^{m}{}_{1}R_{m2}$ is given in Eq. (2)[2]

$$ {}^{m}{}_{1}R_{m2}\,(z,\,\theta) = \begin{bmatrix} cos\theta_z & -sin\theta_z & 0 \\ sin\theta_z & cos\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix}; $$

Similarly${}^{m}{}_{1}R_{m2}$ about y-axis is;

$$ {}^{m}{}_{1}R_{m2}\,(y,\,\theta) = \begin{bmatrix} cos\theta_y & 0 & sin\theta_y \\ 0 & 1 & 0 \\ -sin\theta_y & 0 & cos\theta_y \end{bmatrix}; $$

and ${}^{m}{}_{1}R_{m2}$ about x-axis is;

$$ {}^{m}{}_{1}R_{m2}\,(x,\,\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\theta_x & -sin\theta_x \\ 0 & sin\theta_x & cos\theta_x \end{bmatrix} \tag{2} $$

here, θ_x, θ_y and θ_z represents Euler angles.

Guo et al. [12] obtained closed-form solution for IK of hydraulically driven six d-o-f parallel robot. The problem was solved using Euler angle representation to develop a rotation matrix.

Loop-closure Equation (LE)

Analytical equations are known as loop-closure equations or closed-form equations. This indicates that no numerical processes are used to solve the problem and reduce it to a single-variable polynomial equation [13]. The given equation is solved using functions and mathematical operations for the specified widely recognized set in a closed-form solution. Closed-form solution method eliminates unknown variables from the polynomial equation, resulting in an analytical characteristic polynomial equation with a single variable, which is then solved. Closed-form solutions, in other words, can be stated as analytical expressions. Ji & Wu [14] expressed FK analytically by considering the 3 d-o-f planar PM as two equilateral triangle. As a final solution to this problem, two independent univariate quadratic equations were obtained in a reduced form.

Jacobian Matrix (JM)

To convert joint rates to velocity states, a Jacobian matrix is used. Velocity states and joint rates and are in end-effector space and joint space, respectively. A vector space spanned by the end-effector location is end-effector space, and joint space is a vector space covered by joint variables [2].

Besnard [15] used numerical iterative method based on inverse Jacobian matrix to solve the FK of a 6 d-o-f parallel robot.

Trilateration and Bilateration (T/B)

Trilateration is a formulation that is based on distance. It is used to identify the most feasible position of a point by calculating the known distances from neighboring points to that point. The FK of a 3-RPR PM were addressed by Rojas and Thomas [16], who treated this as equivalent issues, which were defined fully in terms of distances and then trivially resolved through bilateral sequences. There were no exceptions for the formulation stated in terms of distance, while all formulations in terms of poses had exceptions in their derivatives, resulting in numerical instability. Torras et al. [17] solved FK for flagged manipulators by three consecutive trilateration's. These robots belong to a group of PMs such as 3-2-1 and 3/2 manipulators.

Geometric Method (GM)

Simple geometric algorithms can be used to solve kinematics of PM's. Bruyninckx [18] solved the FK using simple geometric algorithms for a fully PM (321-HEXA). Forward position kinematics gives linear and quadratic equations with eight solutions.

Differential Evolution (DE)

The DE algorithm is a problem-solving algorithm for problems with continuous variables. The FK problem has been converted to a problem of optimization and this method also appears to apply in other PMs.

Pond & Carretero [19] used local and global optimization techniques like Quasi-Newton algorithm and Differential Evolution (DE) algorithm to maximize workspace volume for a PRS three degrees-of-freedom (d-o-f) PM.Wang et al. [29] have used an encoded development strategy for global optimization using differential evolution to resolve FK challenges for PM's and the 6-SPS PM [20]. This ensures that the solution is globally optimal and suitable for a variety of PM's and provides a new technology for addressing kinematic problems in the future.

Newton-Raphson Method (NR)

Numerical approaches were used by majority of researchers such as the Newton-Raphson method. The Newton–Raphson approach, often known as Newton's method, is based on linear approximation [21]. This method is named after Joseph Raphson and Isaac Newton.

Newton-Raphson method was used to solve FK of a Stewart-Gough parallel platform [22] and for a spatial 6 d-o-f electro-hydraulic parallel robot [23]. Kim et al. [24] compared numeric and algebraic algorithms for FK and IK of a parallel mechanism (Eclipse). They used numeric method based on Newton-Raphson and obtained closed-form solution.

Interval Analysis (IA)

Interval Analysis, also known as interval computation, is a method for putting a limit on measurement and round-off errors through computerized calculations, resulting in the development of a numerical approach [25].

Merlet [26] solved the FK of a Gough-type parallel robot numerically using interval analysis. This method has several advantages over Newton's scheme, including certified outcomes, reduced processing time, and safety.

Elimination Method (EM)

When a system of polynomial equations is reduced to a single unknown, elimination methods are applied. Sylvester Dialytic Elimination Method is the mostly used method in solving kinematics of robots [2]. Cayley first proposed this approach in 1948. This method simplifies a multi-variable polynomial equation system into a single unknown polynomial equation [2].

Innocenti & Wenger[27] presented of a parallel-actuated spatial manipulator called Tricept.By using the algebraic elimination and back substitution method, 28 solutions in the complex domain were found from a 28th-order univariate algebraic equation, a derivative of a combination of three compatibility equations. Abbasnejadet al. [28] used the Dialytic Elimination Method for the FK of a 4PUS+1PS PM. The equation obtained was then reduced to an eighth-degree polynomial as a single variable. Gallardo-Alvaradoet al. [29] used Sylvester Dialytic elimination method and obtained multiple solutions for the forward position analysis of a 2 (3-RPS) series-PM

Artificial Neural Network (ANN)

Artificial Neural Network (ANN) is a data processing approach that is based on biological neural networks and belongs to the artificial intelligence discipline. ANN is a network of interconnected neurons that receive and transmit data.

Sadjadian et al. [30] found that the FK of a PM was impossible to solve analytically because it involves highly coupled nonlinear equations. Therefore, neural network was used to solve the FK of a three d-o-f actuator redundant hydraulic PM. Finally, it was concluded that ANFIS (Adaptive Neuro-Fuzzy Interference System) provided the best results compared to neural networks. ANFIS is a combination of fuzzy system and neural network.

Fuzzy Inference System (FIS)

The “IF-THEN” rule is used by FIS to anticipate output for uncertain systems. For example, if (temperature) and (humidity) are both high, then (room is hot) [31].

In order to solve the FK of a PM for ankle rehabilitation treatment, Jamwal et al. [32] proposed modified Fuzzy Inference System (FIS). This is a time efficient and very precise method. The gradient descent technology (GD), the genetic Algorithm (GA) and the modified genetic algorithm (MGA) were then used to optimise the FIS. The dynamics of the system are not necessary for GA because it is a model technique. There have been two changes to MGA to improve its performance and local search capabilities. Initially, the best results from the previous generation were individually saved and used for the fitness evaluation of the next generation. Secondly, the speed of convergence was improved. The gradient and the genetic algorithm work together well. GA lacks the ability to search locally, while GD is strong. GA provided a better solution that was closer to the optimal global, but GD

remained stuck in the optimal local position. The results demonstrated that MGA optimises FIS, with respect to accuracy and computation time, exceeding GD, GA, Newton-Raphson and other fuzzy and neural approaches.

TABLE 1. Advantages and disadvantages of each method

Method	Advantages	Disadvantages
DH	Minimum set of scalar parameters	Joint variables (JV) represent relative displacement
ST	JV represent actual displacement	Non minimum set of parameters (2 scalar, 2 vectors)
ANN	Non-algorithm nature of modelling	Long training times and big sizes of networks
EAR	Representation using three Euler angles	Rotation about same axis exhibit singularity
DQ	No singularities occur	Lack of physicality
LE	Better results compared to neural network	Solution is not general, but robot dependent.
DE	It can tackle real and floating-point numbers	Accuracy levels not less than 1^0
JM	Simple computation (numerically robust)	Obtain unpredictable joint configurations
NR	Converge to the real solution in real time	Needs to solve equation systems
T/B	Efficient in terms of location accuracy	Inconsistency of the computation results
IA	Algorithm is numerically robust	Efficiency relies on a set of filtering methods
EM	Obtain all solutions including non-real ones.	Complicated even with the aid of computer algebra
GM	Requires only linear and quadratic equations	Conceptually simple approach but not efficient
FIS	Ability to handle nonlinear systems	Lack of a formal process to define a rule base

CONCLUSION

In robotics, serial manipulators were invented before parallel manipulators. Hence, methodologies for solving forward kinematics were first developed for serial manipulators and many methods are ineffective for parallel manipulators. As a result, new methods for parallel manipulators must be developed, taking into account their unique characteristics as compared to serial manipulators. Different methods for solving forward kinematics of parallel manipulators are presented in this work. The majority of the approaches were also briefly discussed, allowing researchers to choose the most appropriate method and conduct a thorough investigation into their chosen topic. Some of the approaches outlined here can also be used to solve the inverse kinematics of the parallel manipulator and other problems related to parallel manipulators. To deal with the kinematics of the parallel manipulators, the majority of the researchers used the Denavit-Hartenberg Representation, screw theory, and Jacobian Matrix approach. In this paper, all identified methods can be used for solving forward kinematics of parallel and serial manipulators.

ACKNOWLEDGEMENTS

The authors would like to thank Mr. George Joseph, Managing Director (Retd), Rubber Park India (P) Ltd, Ernakulam, for the preparation of the manuscript.

REFERENCES

1. D. Zhang, *Parallel Robotic Machine Tools* (Springer, 2010).
2. L. W. Tsai, *Robot Analysis:The Mechanics of Serial and Parallel Manipulator*(John Wiley & Sons, NewYork, 1999).
3. J. P. Merelet, *Parallel Robots, Solid Mechanics and Its Applications*(Springer, Netherlands, 2006).
4. T. K. Tanev, "Kinematics of a hybrid (parallel-serial) robot manipulator", Mech Mach Theory **35**, 1183-1196 (2000).
5. J. Arata, H. Kondo,N. Ikedo,and H. Fujimoto, "Haptic Device Using a Newly Developed Redundant Parallel Mechanism",IEEE T. Robot.201 – 214(2011).
6. Y. Zhang,H. Liu, andX. Wu, "Kinematics analysis of a novel parallel manipulator",Mech Mach Theory**44**, 1648–1657(2009).

7. J. Duffy, *Statics and Kinematics With Applications to Robotics* (Cambridge University Press, 1996).
8. I. Wolf,and M. Shoham, "Screw theory tools for the synthesis of the geometry of a parallel robot for a given instantaneous task",Mech Mach Theory,656–670(2006).
9. M. Kim,M. S. Han, T. Seo,andT. Lee, "A new instantaneous center analysis methodology for planar closed chains via graphical representation",Int J Control Autom Syst**14**, 1528–1534(2016).
10. J. Zhao, B. Li, X. Yang, and H. Yu, "Geometrical method to determine the reciprocal screws and applications to parallel manipulator", Robotica**27**, 929-940(2009).
11. S. Mustafa, and S. Agrawal, "On the force-closure analysis of n-DOF cable-driven open chains based on reciprocal screw theory",IEEE Trans. Robot**28**, 22-31(2012).
12. H. B. Guo, Y. G. Liu, G. R. Liu, andH. R. Li, "Cascade control of a hydraulically driven 6-DOF parallel robot manipulator based on a sliding mode", Control Eng. Pract.**16**, 1055–1068(2008).
13. H. Bruyninckx, "Closed-Form Forward Position Kinematics for a (3-1-1-1)2 Fully Parallel Manipulator",IEEE Robot. Autom.,326-328(1998).
14. P. Ji, and H. Wu,"An Efficient Approach to the Forward Kinematics of a Planar Parallel Manipulator With Similar Platforms", IEEE Robot. Autom.**18**, 647 – 649(2002).
15. S. Besnard, and W. Khalil, "Self Calibration of Stewart–Gough Parallel Robots Without Extra Sensors",IEEE Robot. Autom.**15**, 1116-1121(1999).
16. N. Rojas,andF. Thomas, "The forward kinematics of 3-RPR planar robots: A review and a distance-based formulation", IEEE Robot. Autom.**27**, 143 – 150(2011).
17. C. Torras, F. Thomas,and M. Alberich-Carramiñana, "Stratifying the Singularity Loci of a Class of Parallel Manipulators", IEEE Robot. Autom.**22**, 23-32(2006).
18. H. Bruyninckx, "The 321-HEXA:A Fully-Parallel Manipulator with Closed-Form Position and Velocity Kinematics,"*Proceedings of the international conference on robotics and automation (1997),* IEEE (1997) New mexico, pp.2657 – 2662.
19. G. Pond, andJ. Carretero, "Singularity analysis and workspace optimization of the inclined PRS manipulator," *Proceedings of the 2004 CSME Forum 2004 CCToMM M^3 Symposium* (2004), pp.1-7.
20. X. Wang, M. Hao, andY. Cheng,"On the use of differential evolution for forward kinematics of parallel manipulators", Appl. Math. Comput.**205**, 760–769(2008).
21. A. Ben-Israel, "A Newton-Raphson Method for the Solution of Systems of Equations" J. Math. Anal. Appl.**15**, 243-252(1966).
22. M. Almonacid, R. J. Saltarén, R. Aracil, andO. Reinoso, "Motion Planning of a Climbing Parallel Rob*ot*", IEEE Robot. Autom.**19**, 485 – 489(2003).
23. C. Yang, Q. Huang,and J. Han, "Decoupling control for spatial six-degree-of-freedom electro-hydraulic parallel robot",Robot Comput Integr Manuf **28**, 14–23(2012).
24. J. Kim, F. C. Park, S. J. Ryu, J. Kim,J. C. Hwang,C. Park, and C. C. Iurascu, "Design and Analysis of a Redundantly Actuated Parallel Mechanism for Rapid Machining",IEEE Robot. Autom.**17**, 423 – 434(2001).
25. D. Degrauwe, G.Lombaert, andG. De-Roeck. "Improving interval analysis in finite element calculations by means of affine arithmetic",Comput Struct **88,**247-254(2010).
26. J. P. Merlet,"Solving the Forward Kinematics of a Gough-Type Parallel Manipulator with Interval Analysis", Int J Robot Res**22**, 221-235(2004).
27. C. Innocenti, andP. Wenger, "Position Analysis of the RRP-3(SS) Multi-Loop Spatial Structure*",* J. Mech. Des.**2**, 272-278(2006).
28. G. Abbasnejad, H. M. Daniali, and A. Fathi*, "*Closed form solution for direct kinematics of a 4PUS + 1PS parallel manipulator*",*Sci.**19**, 320–326(2012).
29. J. Gallardo-Alvarado, C. R. Aguilar-Najera, L. Casique-Rosas, J. M. Rico-Martınez, andM. D. Islam, "Kinematics and dynamics of 2(3-RPS) manipulators by means of screw theory and the principle of virtual work", Mech Mach Theory**43**, 1281–1294(2008).
30. H. Sadjadian, H. D. Taghirad, andA. Fatehi, "Neural networks approaches for computing the forward kinematics of a redundant parallel manipulator", Int. J. Comput. Intelli. **2**, 40-47(2008).
31. M. H. Kazeminezhad, A. Etemad-Shahidi, andS. J. Mousavi, "Application of fuzzy inference system in the prediction of wave parameters", Ocean Eng.**32**, 1709–1725(2005).
32. P. K. Jamwal, S. Q. Xie, Y. H. Tsoi, andK. C. Aw, "Forward kinematics modelling of a parallel ankle rehabilitation robot using modified fuzzy inference",Mech Mach Theory**45**, 1537–1554(2010).